

Retrieval-Augmented Generation (RAG) System Evaluation Report

1. Project Objective

The objective of this project was to design, implement, and evaluate multiple Retrieval-Augmented Generation (RAG) pipelines using different retrieval strategies and Large Language Models (LLMs).

The project compares:

Retrieval Strategies

1. Dense Retrieval
2. Hybrid Retrieval
3. Fusion Retrieval

Language Models

1. GPT4o
2. Llama3
3. Mistral

This resulted in a total of nine RAG pipelines:

- Dense GPT4o
- Hybrid GPT4o
- Fusion GPT4o
- Dense Llama3
- Hybrid Llama3
- Fusion Llama3
- Dense Mistral
- Hybrid Mistral
- Fusion Mistral

The primary goal was to determine which combination of retrieval strategy and language model provides the best balance between retrieval efficiency and answer generation quality.

2. Dataset Preparation

Source Data

The project used approximately 20,000 document chunks stored within a FAISS vector database.

Preprocessing Steps

1. Document Loading
2. Text Cleaning
3. Chunk Creation
4. Embedding Generation
5. FAISS Index Construction

The resulting vector store served as the knowledge base for retrieval.

3. Evaluation Dataset Generation

An evaluation dataset was generated using document summaries.

Each evaluation record contained:

Field	Description
Query	Automatically generated user question
Reference Summary	Ground-truth answer

The evaluation dataset was used to compare generated answers against reference summaries.

4. Retrieval Architectures

4.1 Dense Retrieval

Dense Retrieval uses semantic vector similarity.

Workflow:

User Query

↓

Embedding Generation

↓

FAISS Similarity Search

↓

Top-k Documents

↓

Language Model

Advantages:

- Fast
- Semantically aware
- Low memory usage

Limitations:

- May miss exact keyword matches
-

4.2 Hybrid Retrieval

Hybrid Retrieval combines:

- Dense Retrieval (FAISS)
- Sparse Retrieval (BM25)

Workflow:

Query

↓

Dense Search

+

BM25 Search

↓

Reciprocal Rank Fusion (RRF)

↓

Top Documents

↓

Language Model

Advantages:

- Captures semantic and keyword relevance

Limitations:

- More computationally expensive
-

4.3 Fusion Retrieval

Fusion Retrieval combines:

- Dense Retrieval
- Sparse Retrieval
- Web Search Retrieval

Workflow:

Query

↓

Dense Retrieval

+

Sparse Retrieval

+

Web Retrieval

↓

Merged Context

↓

Language Model

Advantages:

- Largest information coverage

Limitations:

- Increased latency
 - Increased memory consumption
 - Potential retrieval noise
-

5. Evaluated Pipelines

The following pipelines were implemented:

Retrieval	GPT4o	Llama3	Mistral
Dense	✓	✓	✓
Hybrid	✓	✓	✓
Fusion	✓	✓	✓

Total Pipelines Evaluated:

9

6. Generation Evaluation Metrics

Three standard Natural Language Generation (NLG) metrics were used.

ROUGE-L

Measures overlap between generated answers and reference summaries using the Longest Common Subsequence (LCS).

Desired Value:

Higher

BLEU

Measures n-gram similarity between generated and reference text.

Desired Value:

Higher

BERTScore

Measures semantic similarity using contextual embeddings.

Desired Value:

Higher

7. Generation Results

Rank	Pipeline	ROUGE-L	BLEU	BERTScore
1	Fusion GPT4o	0.2977	0.0476	0.8588
2	Hybrid Mistral	0.2712	0.0383	0.8515
3	Fusion Mistral	0.2690	0.0393	0.8498
4	Dense GPT4o	0.2385	0.0316	0.8474
5	Dense Mistral	0.2432	0.0226	0.8472
6	Fusion Llama3	0.2430	0.0353	0.8453
7	Hybrid Llama3	0.2389	0.0379	0.8433
8	Hybrid GPT4o	0.1915	0.0136	0.8419
9	Dense Llama3	0.2373	0.0287	0.8387

8. Retrieval and Runtime Metrics

The following operational metrics were collected:

- Latency
- Memory Usage
- Context Length
- Response Length
- Context Relevance

9. Retrieval Results

Pipeline	Latency	Memory	Context Length	Response Length	Context Relevance
Dense GPT4o	2.37	0.03	2143	674	0.5622
Dense Llama3	27.12	46.06	2143	702	0.5622

Pipeline	Latency	Memory	Context Length	Response Length	Context Relevance
Dense Mistral	34.92	6.81	2143	668	0.5615
Fusion GPT4o	3.09	75.11	4542	735	0.5417
Fusion Llama3	43.53	0*	4616	947	0.5390
Fusion Mistral	55.97	0*	4714	791	0.5464
Hybrid GPT4o	2.08	8.35	2226	551	0.5587
Hybrid Llama3	26.12	24.37	2226	824	0.5587
Hybrid Mistral	37.48	0*	2226	680	0.5589

*Negative memory measurements observed during experimentation were attributed to garbage collection and operating system memory reclamation and were treated as zero during analysis.

10. Key Findings

Best Generation Pipeline

Fusion GPT4o achieved the highest:

- ROUGE-L
- BLEU
- BERTScore

This indicates superior lexical overlap and semantic similarity with the reference summaries.

Most Efficient Retrieval Pipeline

Dense GPT4o achieved:

- Highest Context Relevance
- Lowest Memory Consumption
- Very Low Latency

This makes Dense GPT4o the most efficient retrieval pipeline.

Fastest Pipeline

Hybrid GPT4o achieved the lowest latency:

2.08 seconds

making it the fastest pipeline.

Largest Information Coverage

Fusion Mistral produced the largest context length:

4714 characters

indicating maximum retrieval coverage.

Longest Responses

Fusion Llama3 produced the longest responses:

947 characters on average.

11. Precision, Recall, and MRR Limitation

Traditional retrieval metrics such as:

- Precision
- Recall
- Mean Reciprocal Rank (MRR)

could not be computed reliably because the retrieval datasets did not contain explicit relevance labels indicating whether retrieved documents were relevant to each query.

Without ground-truth relevance judgments:

- Precision cannot be calculated.
- Recall cannot be calculated.
- MRR cannot be calculated.

Therefore, evaluation focused on generation metrics and operational retrieval metrics.

12. Final Interpretation

An important observation emerged during experimentation.

Dense GPT4o retrieved the most relevant context.

However,

Fusion GPT4o generated the highest-quality answers.

This suggests that:

Retrieval Quality Alone $\neq$ Answer Quality

Instead,

Retrieval Coverage + Model Capability

plays a crucial role in answer generation performance.

Although Fusion Retrieval introduced slightly more retrieval noise, the additional contextual information enabled GPT4o to generate more accurate and semantically richer responses.

13. Final Conclusion

The experimental results demonstrate that retrieval strategy significantly impacts the effectiveness of Retrieval-Augmented Generation systems.

Dense Retrieval provided the most efficient retrieval performance.

Hybrid Retrieval provided the fastest response times.

Fusion Retrieval provided the richest contextual information.

Among all evaluated pipelines, Fusion GPT4o achieved the highest ROUGE-L, BLEU, and BERTScore values, indicating superior answer quality and semantic correctness.

Therefore, Fusion GPT4o is selected as the best overall pipeline and the final recommended architecture for this project.

Final Rankings

Best Overall Pipeline

Fusion GPT4o

Best Retrieval Efficiency

Dense GPT4o

Fastest Pipeline

Hybrid GPT4o

Largest Context Coverage

Fusion Mistral

Longest Generated Responses

Fusion Llama3

Best Semantic Similarity

Fusion GPT4o

Recommended Production Architecture

Fusion Retrieval + GPT4o